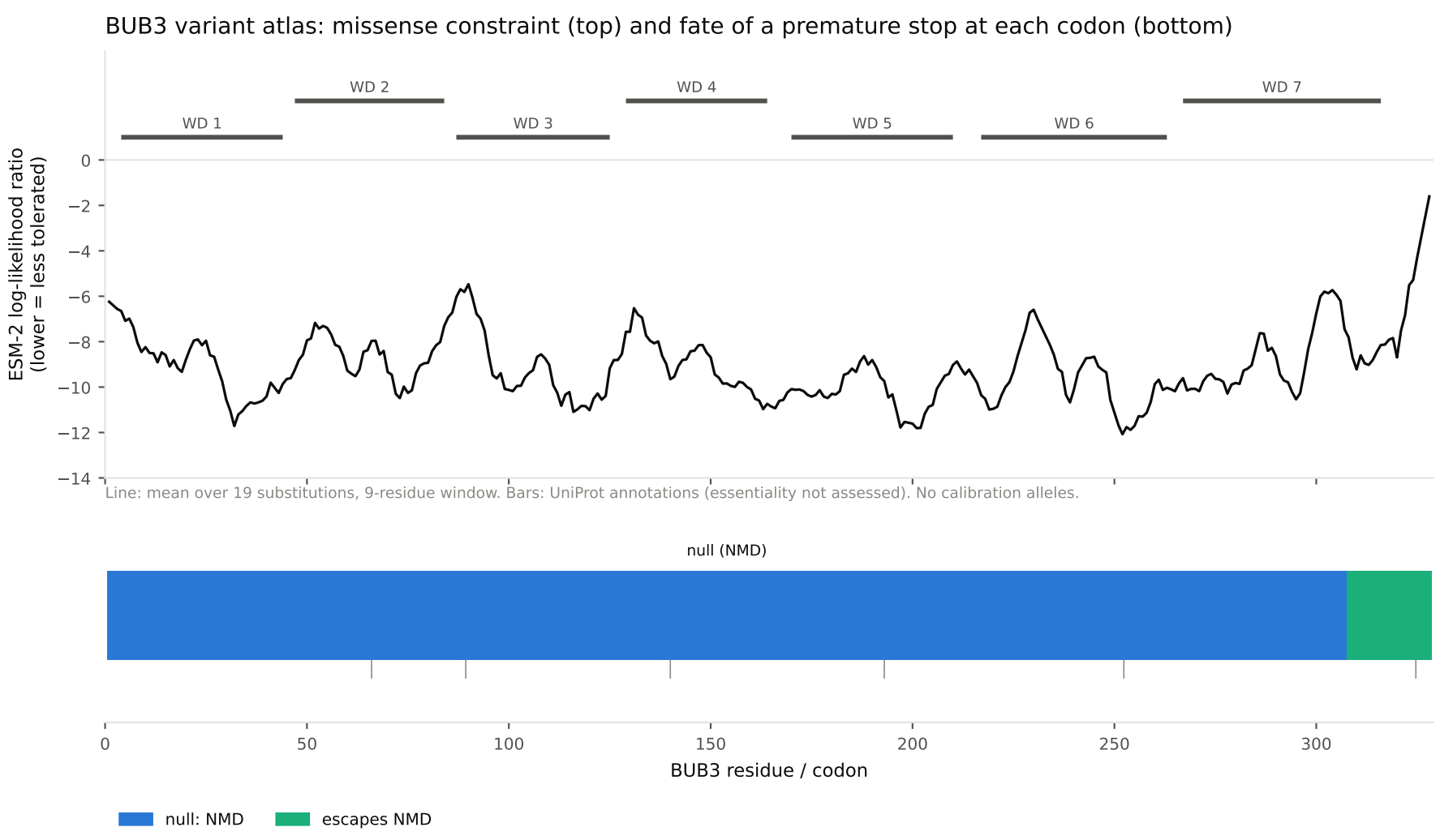
BUB3 variant atlas: missense constraint (top) and fate of a premature stop at each codon (bottom)
WD 1
WD 2
WD 3
WD 4
WD 5
WD 6
WD 7
ESM-2 log-likelihood ratio
(lower = less tolerated)
0
−2
−4
−6
−8
−10
−12
−14
Line: mean over 19 substitutions, 9-residue window. Bars: UniProt annotations (essentiality not assessed). No calibration alleles.
null (NMD)
0
50
100
150
200
250
300
BUB3 residue / codon
null: NMD
escapes NMD